INVERTIS UNIVERSITY BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY • DEPARTMENT KNOWLEDGE ARCHIVES

B.Tech Final-Year Capstone Project Guidelines & Manual

Authority: Department of Computer Science & Engineering, Academic Council | Regulation: Session 2023-2024 | Curriculum Pillar: Phase 5

1. PROJECT LIFECYCLE & MILESTONE PHASES

- Phase I: Project Selection & IEEE Synopsis Defense (Within first 3 weeks).
- Phase II: Architecture Specification, SRS Documentation & Mid-Term Review (Week 7).
- Phase III: Prototype Implementation, Benchmarking & Code Verification (Week 12).
- Phase IV: Final Technical Dissertation Submission & External Viva Voce (Week 15).

2. APPROVED TECHNICAL DOMAINS

- Artificial Intelligence & Machine Learning: Explainable AI, LLM RAG pipelines, Vision.
- Distributed Systems & Cloud Computing: Microservices, Kubernetes, High-availability APIs.
- Cyber Security & Cryptography: Zero-trust architectures, vulnerability scanners.
- Internet of Things (IoT) & Embedded Robotics: Autonomous navigation, sensor telemetry.

3. EVALUATION RUBRIC & MARKS DISTRIBUTION (TOTAL: 100 MARKS)

- Literature Review & Problem Formulation: 20 Marks (Internal Guide).
- System Architecture & Innovative Engineering: 30 Marks (Department Committee).
- Working Implementation & Quality of Code: 30 Marks (Lab Demonstration).
- Viva Voce Defense & Technical Report Presentation: 20 Marks (External Examiner).